

CHAPTER 1

SOFTWARE PROJECT MANAGEMENT PLAN

1.1 Introduction

Quaid-i-Azam University provides hostel accommodation to students who come from different cities and require residence near the university campus. Managing student accommodation involves several administrative activities, including hostel applications, room allotment, payment verification, complaint handling and maintenance records.

The proposed **Hostel Management System for Quaid-i-Azam University** is a web-based application designed to organize these activities through a centralized platform. The system will allow students to apply for hostel accommodation, check their application status, submit complaints and view notices. Hostel staff and wardens will use the system to review applications, manage rooms, allocate beds and generate reports.

The purpose of the system is to reduce paperwork, prevent record duplication and improve communication between students and hostel administration.

1.2 Problem Statement

Hostel management becomes difficult when records are maintained manually or across separate files. Staff members may need to search through multiple records to check room availability, student information or payment status. Students may also need to visit the hostel office repeatedly to ask about their applications.

Common problems include:

- Delays in application processing
- Difficulty in tracking vacant rooms and beds
- Duplicate room allotments
- Incomplete student records
- Poor complaint tracking
- Difficulty in verifying payments
- Limited access to reports
- Repeated office visits by students
- Loss of historical allotment information

A centralized Hostel Management System is required to solve these problems and provide accurate information to authorized users.

1.3 Project Description

The Hostel Management System is a web application that manages hostel-related activities at Quaid-i-Azam University. It stores student records, hostel information, room details, bed availability, applications, allotments, complaints, fee records and notices.

Students will use the system to:

- Submit hostel applications
- Apply for renewal
- Track application status
- View allotment details
- Submit complaints
- Check complaint status
- View fee status
- Read hostel notices

Hostel staff and wardens will use the system to:

- Review applications
- Approve or reject requests
- Add hostel and room details
- Allocate beds
- Manage renewals
- Verify fee records
- Handle complaints
- Generate reports

The system will use role-based access control. Each user will only access the features related to their responsibilities.

1.6 Major Functionalities

1.6.1 User Authentication

The system will provide secure login functionality. Students, wardens, staff and administrators will use separate accounts. The system will verify user credentials and open the correct dashboard.

1.6.2 Hostel Application

Students will submit hostel applications online. The application will contain personal, academic and contact information. After submission, the system will generate a tracking number.

1.6.3 Application Review

Hostel staff will review submitted applications. They will verify student information and update the application status. An application may be approved, rejected, returned for correction or placed on a waiting list.

1.6.4 Hostel and Room Management

Authorized staff will add hostel buildings, rooms and beds. Each bed will have a status, such as available, occupied, reserved or unavailable.

1.6.5 Room Allotment

After approval, an available bed will be assigned to the student. The system will update the bed status automatically and save the allotment record.

1.6.6 Renewal Request

Existing residents will submit renewal requests for the next semester or academic session. Staff members will review the requests and update their status.

1.6.7 Complaint Management

Students will submit complaints related to water, electricity, furniture, internet, sanitation or maintenance. Staff members will update the complaint status until the issue is resolved.

1.6.8 Fee Management

The system will maintain student hostel-fee records. Staff members will verify payments and update payment status.

1.6.9 Notice Management

Wardens and authorized staff will publish important notices, such as deadlines, fee reminders and maintenance updates.

1.6.10 Reports

The system will generate reports related to:

- Occupied beds
- Available beds
- Pending applications
- Approved applications
- Renewal requests
- Complaint status
- Payment status
- Visitor records

1.7 Users of the System

User	Responsibilities
Student	Submit applications, track status, view allotment, submit complaints and read notices
Hostel Office Staff	Review applications, verify records, manage payments and generate reports
Warden	Approve requests, allocate rooms, manage complaints and publish notices
Security Staff	Record visitor entry and exit
System Administrator	Manage users, roles, settings and backups
University Management	View summary reports and statistics
